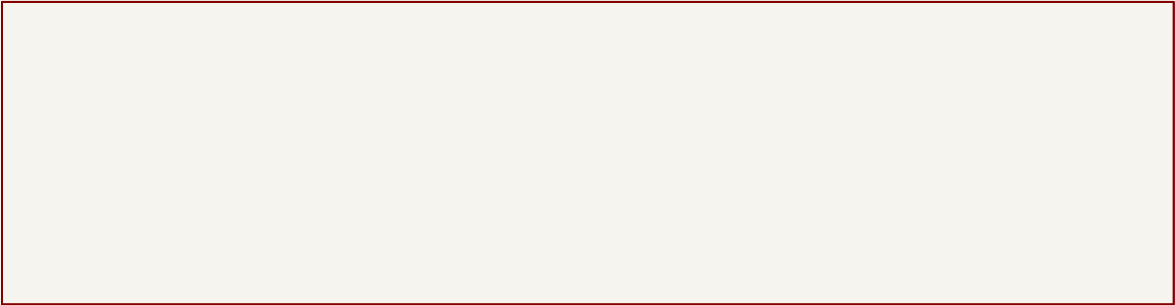


EDITABLE CIRCUIT CAPTURE C1 / not routed, fabricated or bench qualified

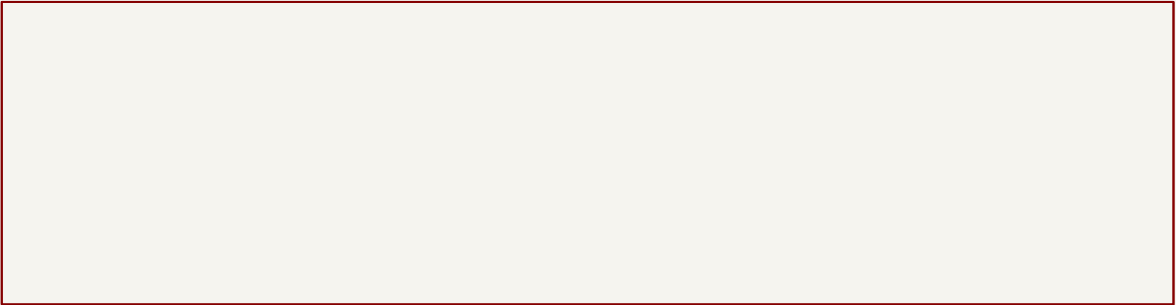
Every semiconductor pin, fitted passive and connector is explicit. NC pins have native no-connect flags.

PD Input, eFuse and remote regulators



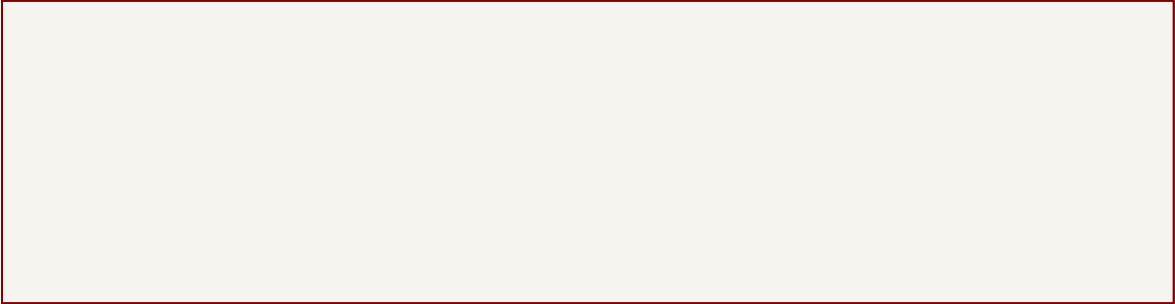
File: power.kicad_sch

Fan/servo switches, PWM and feedback connectors



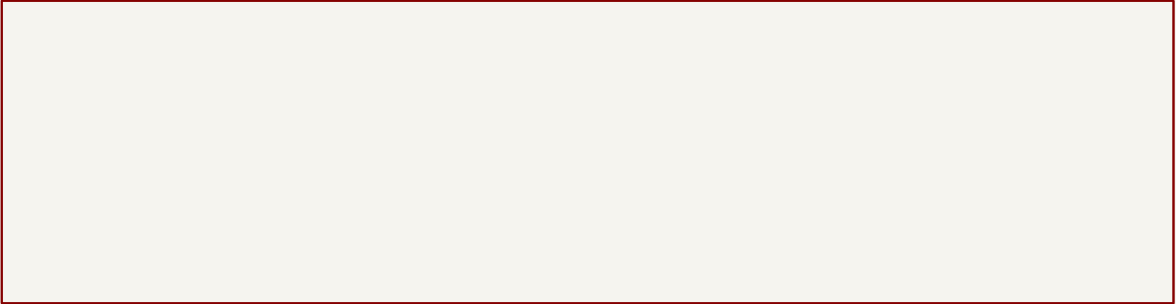
File: fan_servo.kicad_sch

Downward ambient lighting supply and data



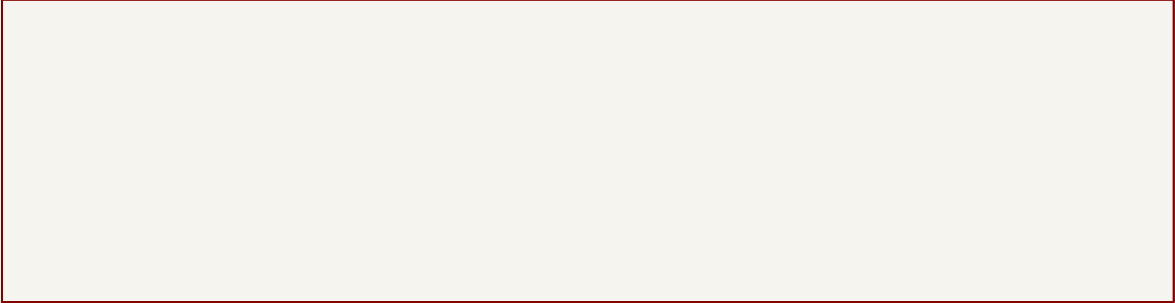
File: ambient.kicad_sch

Encoder, I2C, grille safety and remote Pico



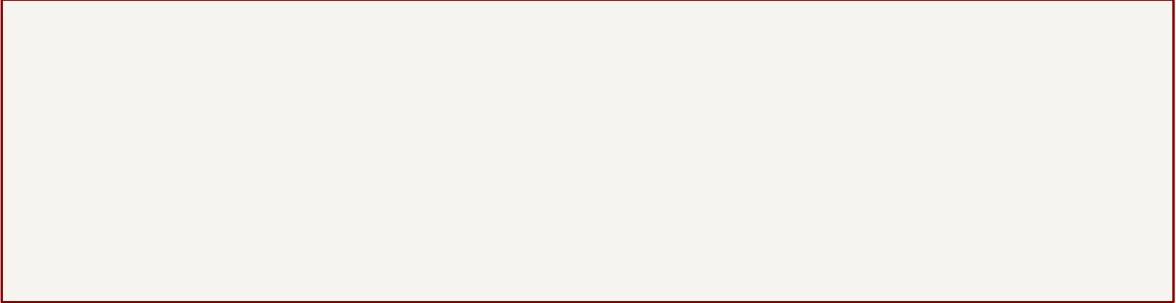
File: Inputs.kicad_sch

ADC Isolation and voltage/feedback measurement



File: adc.kicad_sch

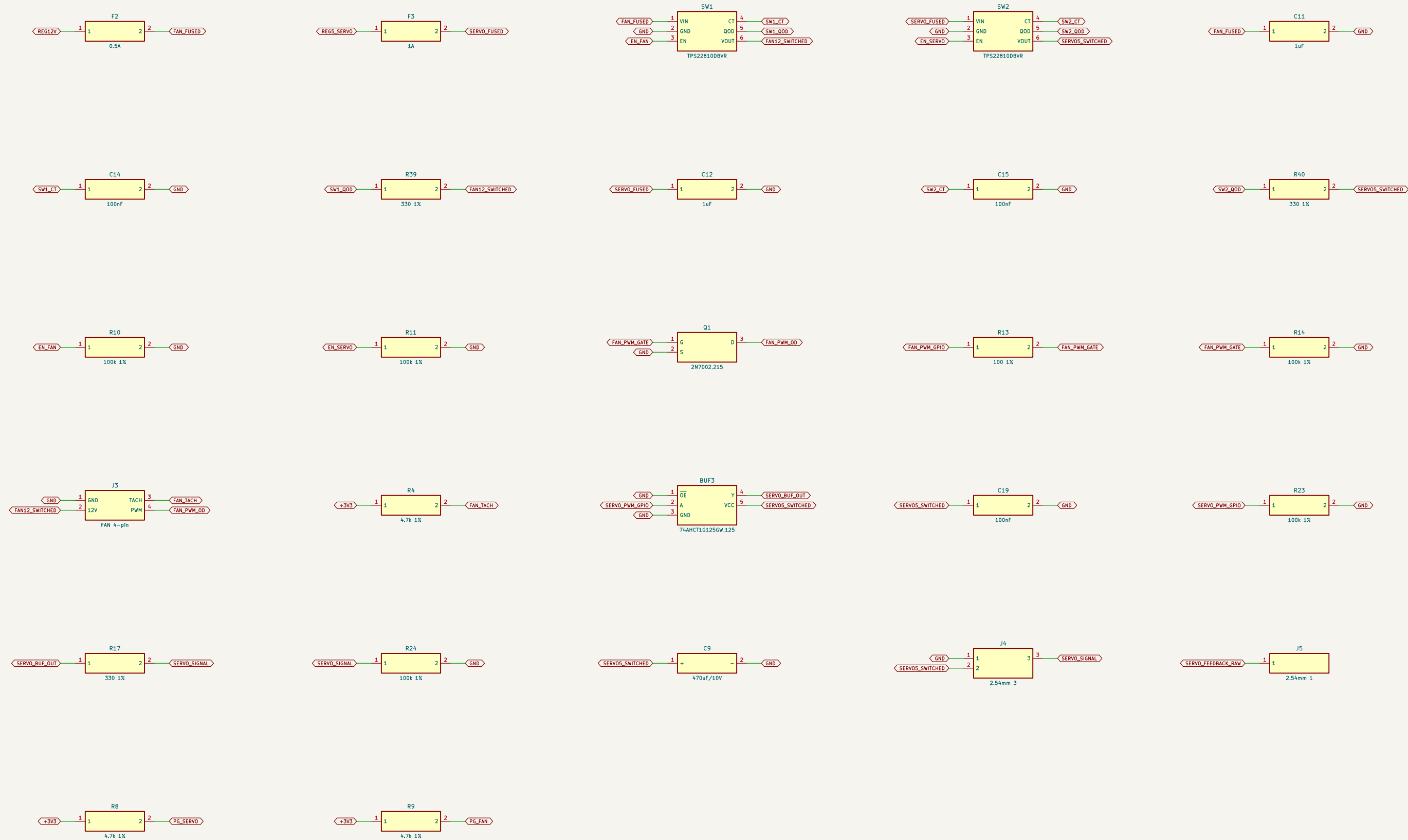
IPS display interfaces and external screen passives



File: screen.kicad_sch



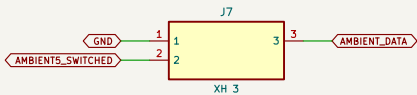
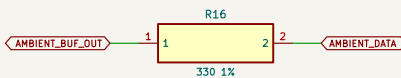
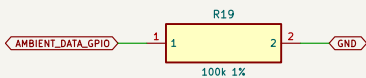
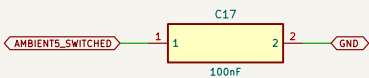
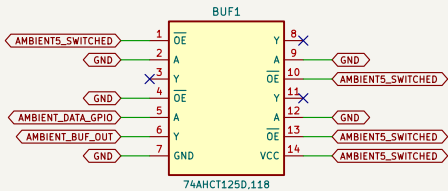
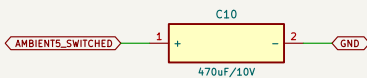
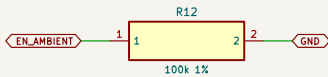
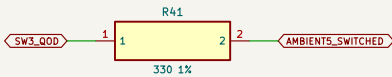
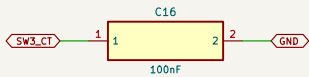
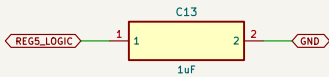
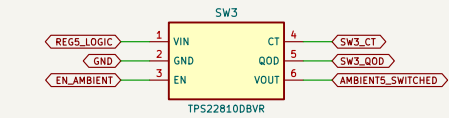
Board allowance90x65mm; component placement awaits pressure hose and harness keepouts.



FanPWM uses Q1 open-drain inversion; no12V pullup. EN pulldowns are mandatory.

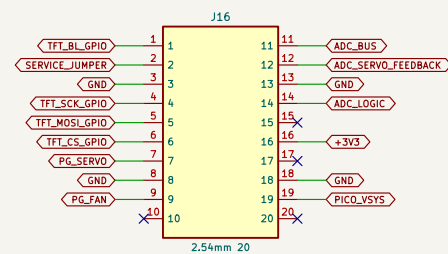
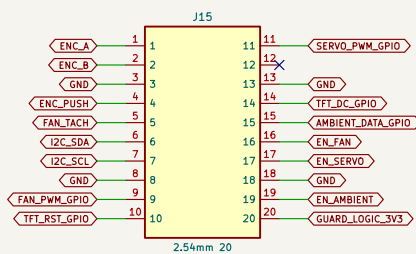
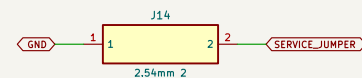
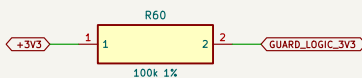
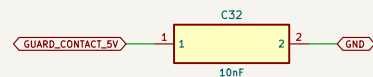
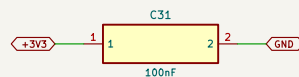
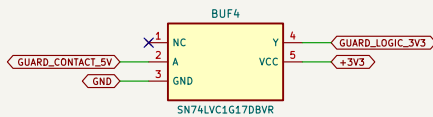
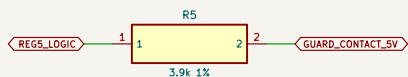
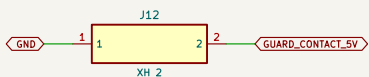
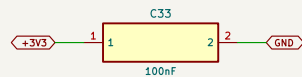
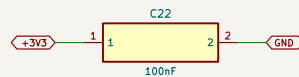
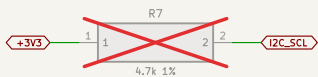
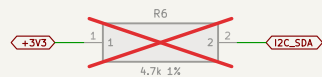
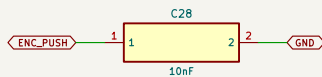
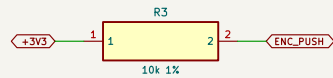
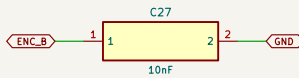
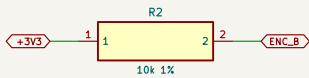
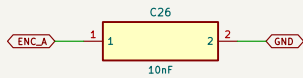
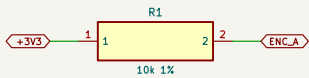
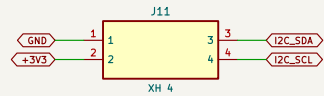
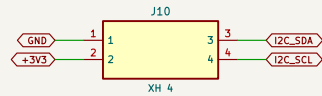
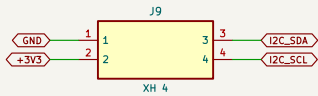
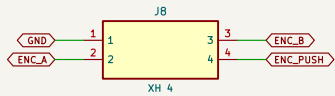
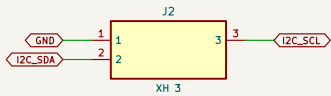
BUF3 is powered only by switched servo supply. 330ohm QOD resistors are1206/0.25W.

Module PG is open-drain; pullups go only to3.3V. Fuse sizes do not replace timed servo jam cutoff.



BUF1 supply follows switched ambient rail. Its three unused channels are disabled and inputs grounded.

The retired main pixel bars/status lamp are absent. The 8-pixel ambient stick is the only LED chain.

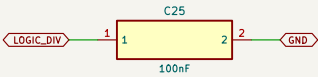
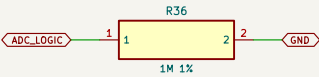
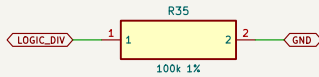
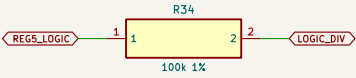
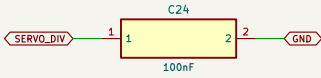
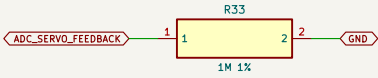
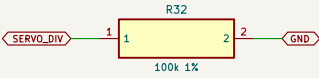
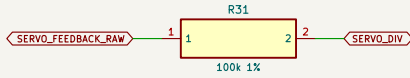
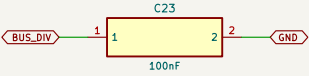
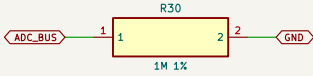
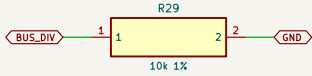
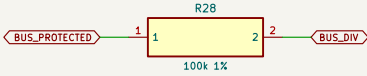
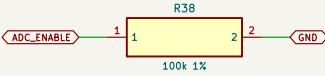
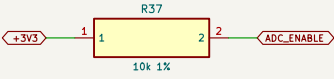
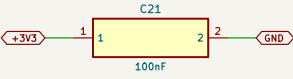
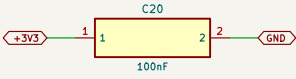
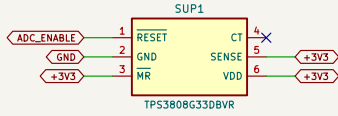
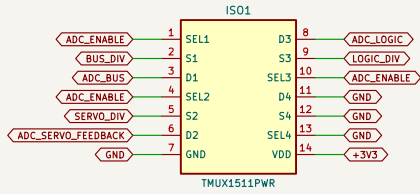


BUF4 translates 5V grille-contact node to 3.3V GP15; keep raw guard net away from Pico.

R6/R7 are DNP until combined breakout pullups are measured. Sensor/temperature supplies are 3.3V.

J15contact12 / J16contact10 are physically removed and cable keyed; Pico35/37/40 are NC.

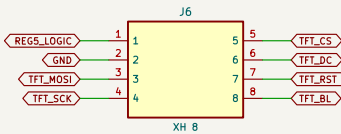
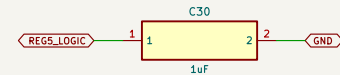
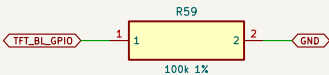
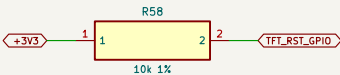
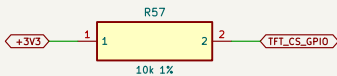
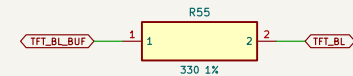
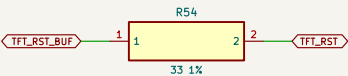
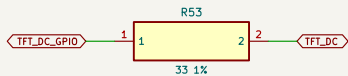
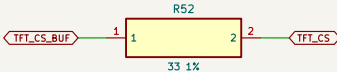
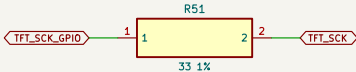
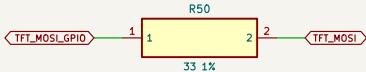
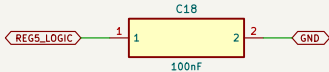
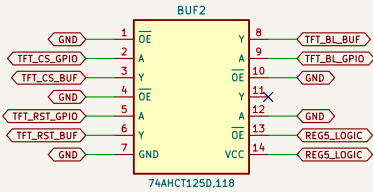
J15positionn=Pico physcaln; J16positionn=Pico physcaln+20. Module orientation does not change numbering.



Filter capacitors are before ISO1. ADC-side 1M pulldowns remain after it; no stored capacitor on GPIO side.

SUP1 CT is NC for nominal 20ms delay. /RESET enables ISO1 only after 3.3V is healthy.

Bus nominal scale 11.1; logic/servo network 2.1. Servo uses measured raw feedback table.



BUF2 is unswitched5V. CS/RST are buffered because module pullups go toVIN; GPIO must not see5V.

R56 and C30 are fitted at the display end and excluded from carrier PCB population.

GP6fan, GP8servo and GP16BL are different PWM slices. Keep combined Pico/display wire length under150mm.